

ECE 445
SENIOR DESIGN LABORATORY
FINAL REPORT

**Voice-Controlled FPV Drone via
ExpressLRS**

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Abstract

This report presents the design and verification of a voice-controlled quadrotor drone system. The ground unit consists of an ESP32 microcontroller paired with a VC02 offline voice recognition module and an SSD1306 OLED display. Seven spoken commands—takeoff, land, forward, backward, left, right, and stop—are mapped to pre-defined RC channel sequences encoded as CRSF frames and transmitted at 50 Hz to a RadioMaster RP2 module reflashed with ExpressLRS transmitter firmware. The RP2 bridges the commands over a 2.4 GHz RF link to a RadioMaster RX24T receiver on the aircraft, which forwards decoded channels to a SpeedyBee F405 V5 flight controller running Betaflight. Four 1860 KV brushless motors driven by an OX32 55 A ESC execute the commanded maneuvers. Barometric altitude telemetry is returned over the same RF link and displayed on the ground unit's OLED in real time.

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1 Introduction

1.1 Motivation

Unmanned aerial vehicles (UAVs) controlled through conventional radio transmitters require the operator to hold a dedicated handheld unit with two joysticks throughout a flight session. This form factor demands bilateral manual dexterity, prevents the operator from simultaneously performing other tasks, and creates a barrier for users without prior RC flight experience. Hands-free or minimal-hardware control interfaces could lower this barrier and enable use cases in which the operator’s hands must remain free, such as first-response inspection or assistive applications.

Voice command is a well-understood interaction modality that requires no additional hardware beyond a microphone and a recognition module, imposes no constraint on hand posture, and maps naturally to a discrete set of high-level flight maneuvers. This project investigates whether an offline voice-command ground unit can provide sufficient control fidelity for stable takeoff, directional flight, and landing of a five-inch class quadrotor, using the ExpressLRS open-source RF link as the wireless bridge.

1.2 Background and Related Work

1.2.1 FPV Quadrotor Platform

Five-inch class FPV (first-person-view) quadrotors are a well-characterized platform that balances agility, payload capacity, and component availability. Standard flight controllers in this class run the Betaflight open-source firmware, which exposes a CRSF serial interface for RC input and returns sensor telemetry on the same link. This existing ecosystem provides a stable, well-documented integration target for the ground control unit.

1.2.2 ExpressLRS

ExpressLRS (ELRS) is an open-source 2.4 GHz RC link protocol derived from the LoRa physical layer [1]. It achieves packet update rates from 50 Hz to 1000 Hz with measured round-trip latencies below 5 ms at 250 Hz, significantly lower than legacy protocols such as Futaba S-FHSS (55 Hz) or FrSky D16 (150 Hz). The open firmware stack allows the protocol to run on commodity hardware including the RadioMaster RP1 and RP2 modules, which use an Espressif ESP8285 microcontroller and a Semtech SX1280 2.4 GHz transceiver.

1.2.3 Offline Voice Recognition

Offline keyword-spotting modules such as the VC02 perform fixed-vocabulary recognition entirely on-device without network connectivity, achieving recognition latencies typically below 500 ms and operating reliably in environments where cloud-based services are impractical. The VC02 communicates recognized commands as single-byte UART messages, making integration with a microcontroller straightforward.

1.3 System Overview

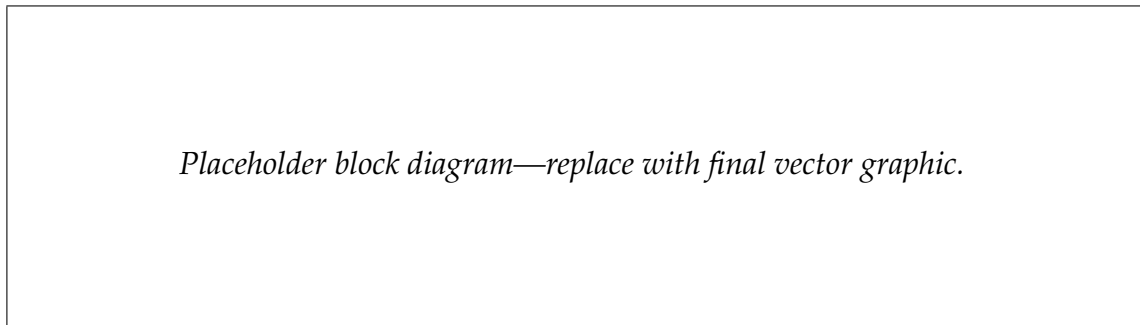


Figure 1: Voice-controlled FPV drone system architecture (ground unit, RF link, and airframe).

Figure 1 shows the complete system. The ground unit consists of an ESP32 connected to a VC02 voice recognition module and an SSD1306 OLED display. The ESP32 encodes operator voice commands as CRSF RC frames and drives the RadioMaster RP2 module via UART. The RP2, reflashed with ELRS transmitter firmware, broadcasts RC frames at 50 Hz over the 2.4 GHz ISM band. On the aircraft, a RadioMaster RX24T ELRS receiver forwards decoded RC channel data over CRSF to the SpeedyBee F405 V5 flight controller running Betaflight. The flight controller drives four 1860 KV brushless motors through an OX32 55 A four-in-one ESC. Barometric altitude telemetry from the flight controller is relayed back through the same RF link and decoded by the ESP32, then displayed on the OLED for the ground operator.

1.4 High-Level Requirements

Table 1 lists the top-level performance requirements for the system.

Table 1: High-level system requirements.

Requirement	Description	Target
R1	Voice command recognition latency from utterance end to VC02 UART byte	<1 s
R2	End-to-end command latency from VC02 byte to CRSF frame departure at RP2	<40 ms
R3	RF link update rate during flight	≥ 50 Hz
R4	Stable hover after takeoff command without pilot intervention	≥ 5 s
R5	Directional movement response to forward/backward command	Observable displacement
R6	Successful autonomous landing (disarm without crash)	100% of attempts
R7	Barometric altitude displayed on OLED during flight	Continuous

2 Design

This chapter presents the design of each subsystem of the voice-controlled drone system. As shown in Figure 1, the system comprises three subsystems: the voice control unit, which converts spoken commands into CRSF RC frames; the ExpressLRS (ELRS) RF communication link, which carries those frames wirelessly to the aircraft; and the drone flight system, which executes the commands. The following sections describe the design procedure and implementation details for each subsystem.

2.1 Voice Control Subsystem

2.1.1 Design Procedure

The ground-side control subsystem must convert operator intent into RC channel values compatible with the CRSF protocol and transmit them to the airborne flight controller at a sufficient rate to maintain a stable RF link. Two interaction paradigms were evaluated: continuous gesture-based input derived from inertial sensors, and discrete voice command recognition.

Gesture-based input with an IMU offers proportional control authority but requires the operator to maintain deliberate wrist posture throughout flight, increasing physical fatigue and the risk of inadvertent inputs. Voice command recognition maps a fixed vocabulary of spoken words to pre-defined flight maneuvers, reducing the physical interface to speech and freeing the operator's hands. For a proof-of-concept demonstration in which the primary goal is autonomous takeoff, directional movement, and landing, a discrete command set is sufficient and voice recognition was adopted.

2.1.2 Hardware

The control unit is built on an ESP32 microcontroller module, selected for its dual hardware UART peripherals, 240 MHz Xtensa LX6 core, and broad Arduino ecosystem support. A VC02 offline voice recognition module is connected to the ESP32 on UART2 (ESP32 GPIO 16 RX, GPIO 17 TX) at 115,200 bps. The VC02 performs on-device keyword spotting without a cloud connection, returning a single command byte over UART each time a recognized phrase is detected. A 128×64 pixel OLED display (SSD1306, I²C address 0x3C) is connected on the I²C bus and shows the current action state, barometric altitude, and CRSF link status to the ground operator in real time.

2.1.3 Voice Command Set and Channel Mapping

The VC02 is configured to recognize seven command words. Table 2 lists each command, the byte value returned by the VC02, and the resulting RC channel action produced by the firmware.

All channel values are in standard RC pulse-width units (μ s). The neutral (mid-stick) value is 1500 μ s and the arm/disarm channel (CH5) is toggled between 1000 μ s (disarmed) and 2000 μ s (armed). Commands 3–7 are accepted only when the aircraft is in

Table 2: Voice command set and corresponding flight actions.

VC02 byte	Command	Flight action
1	Takeoff	Arm motors, ramp throttle: 1250 μ s for 1.5 s, then 1180 μ s for 1.5 s, settle at hover throttle 1100 μ s
2	Land	Reduce throttle gradually to 1050 μ s over 2 s, then disarm
3	Forward	Pitch channel to 1550 μ s (forward tilt) for 1200 ms, return to neutral
4	Backward	Pitch channel to 1450 μ s (backward tilt) for 1200 ms, return to neutral
5	Left	Yaw channel held at 1450 μ s until next command
6	Right	Yaw channel held at 1550 μ s until next command
7	Stop / Brake	If moving forward: reverse pitch pulse for braking; if moving backward: forward pitch pulse; otherwise: hold hover

the flying state; any such command issued before takeoff is silently discarded to prevent unintended ground arming.

2.1.4 CRSF Frame Generation and Telemetry

The ESP32 encodes 16 RC channels into CRSF frames and transmits them to the RadioMaster RP2 on UART1 (ESP32 GPIO 25 TX, GPIO 26 RX) at 115,200 bps. Each 11-bit channel value is linearly mapped from the 1000–2000 μ s range to the CRSF integer range 192–1791 using the relation

$$v_{\text{CRSF}} = \frac{(v_{\mu\text{s}} - 1000) \times 1600}{1000} + 192 \quad (1)$$

The 16 values are then packed into 22 bytes with 11 bits per channel (176 bits total), appended with a DVB-S2 CRC-8 byte computed over the frame type and payload fields, and wrapped in the 26-byte CRSF envelope (address 0xC8, length byte, type 0x16). Frames are sent every 20 ms (50 Hz), the minimum rate required to prevent the ELRS receiver from triggering a link-loss failsafe.

On the downlink, the firmware parses incoming CRSF telemetry frames using a byte-at-a-time state machine that resets whenever a gap of more than 5 ms is detected between consecutive bytes, providing robustness against partial frames caused by UART contention. Barometric altitude telemetry (CRSF frame type 0x09) is decoded and displayed on the OLED in real time, giving the operator situational awareness without a separate ground station laptop.

2.2 ELRS RF Communication Subsystem

2.2.1 Design Procedure

Three 2.4 GHz RC link protocols were evaluated: Futaba S-FHSS, FrSky ACCESS, and ExpressLRS (ELRS). Table 3 summarizes the key characteristics of each.

Table 3: Comparison of evaluated 2.4 GHz RC link protocols.

Protocol	Max update rate	Min latency	Open source	TX power
Futaba S-FHSS	55 Hz	≈ 18 ms	No	Fixed
FrSky ACCESS	150 Hz	≈ 7 ms	No	Fixed
ELRS 2.4 GHz	1000 Hz	< 5 ms	Yes	Configurable

Futaba S-FHSS operates at a fixed 55 Hz and uses a closed firmware ecosystem, precluding integration with custom transmitter hardware. FrSky ACCESS supports up to 150 Hz but similarly relies on proprietary firmware. ELRS is an open-source protocol supporting update rates from 50 Hz to 1000 Hz with a publicly documented firmware stack that can be compiled and deployed on commodity hardware. ELRS was selected for its low latency, configurable update rate, and compatibility with custom transmitter hardware.

For the transmitter unit, two hardware options were considered. The first is a dedicated ELRS TX module (e.g., RadioMaster Ranger Micro), which provides a self-contained solution with a preconfigured USB interface. The second is a RadioMaster RP2 receiver module reflashed with ELRS transmitter firmware. A dedicated TX module adds unnecessary cost and physical bulk for a fixed ground station installation. The RP2 module, when flashed with the *RX as TX* firmware variant, performs identically as a transmitter while occupying a smaller footprint at lower cost. The RP2-as-TX approach was adopted.

2.2.2 Hardware

The RadioMaster RP2 module contains two primary integrated circuits. The Espressif ESP8285 is a single-chip microcontroller integrating a 32-bit Tensilica L106 core and a 2.4 GHz 802.11b/g/n Wi-Fi radio. In the ELRS firmware context, the ESP8285 executes the protocol stack, manages the CRSF serial interface to the host device, and controls the RF transceiver over SPI. The Semtech SX1280 is a 2.4 GHz multi-mode transceiver supporting FLRC (Fast Long-Range Communication) modulation, which ELRS employs at high update rates to achieve packet air times below 1 ms and end-to-end round-trip latencies below 5 ms at 250 Hz.

2.2.3 Firmware Configuration

Firmware was compiled using the ExpressLRS Configurator application. Three parameters were fixed at compile time.

Firmware target. The target `Generic 2.4 GHz /Generic ESP8285 2.4 GHz RX as TX` was selected. This variant configures the ELRS firmware to initiate and maintain the RF link, assigning the module the transmitter role in the ELRS packet timing scheme. A standard receiver build (`Generic ESP8285 2.4 GHz RX`) causes the module to listen for an external transmitter signal and never initiate packets, making it unsuitable for ground-side use.

Regulatory domain. The `ISM_2400` domain was selected in preference to `CE_LBT` (Listen-Before-Talk). The `CE_LBT` domain mandates that the transmitter sense channel occupancy before each transmission and defer if the channel is busy. This listen phase adds a variable pre-transmission delay and reduces the time available for telemetry data within each packet interval. At high update rates (500 Hz and 1000 Hz), the cumulative LBT overhead measurably reduces telemetry throughput. The `ISM_2400` domain removes this restriction, restoring full packet capacity for combined command and telemetry data.

Binding Phrase. A user-defined alphanumeric string is compiled into both the transmitter and receiver firmware builds. At power-on, any two ELRS devices sharing an identical Binding Phrase and a compatible major firmware version synchronize automatically, without a physical bind-button pairing sequence. This eliminates the multi-step binding procedure required by traditional RC protocols and reduces the risk of inadvertent binding to a third-party transmitter.

2.2.4 Initial Firmware Flashing

The ESP8285 is programmed via UART using the flashing tool integrated into the ExpressLRS Configurator. The chip enters its UART bootloader only when the GPIO0 strap-ping pin is sampled low at power-on. The prescribed procedure is: (1) hold the RP2's Boot button, asserting GPIO0 low through the onboard pull-down circuit; (2) apply power to the module; (3) release the button. The configurator then transfers the firmware image over UART at 921,600 baud.

A hardware behavior present on some RP2 units causes the module to enter bootloader mode automatically when a USB-to-UART adapter is connected before power is applied. During the pre-enumeration interval, certain adapters drive their TX and RX lines low rather than holding the idle-mark (logic high) state. The adapter may hold its TX line low during the reset window. That state is read as a bootloader entry request on GPIO0. The recommended mitigation is to supply power to the RP2 first, allow it to complete its normal boot sequence, then connect the adapter's TX and RX lines. This ensures the strap-ping pins are sampled in the normal-boot state before the adapter's lines are driven.

After the initial UART flash, subsequent firmware updates can be applied over Wi-Fi through the module's built-in web interface (WebUI), eliminating the need for the Boot button procedure on future updates.

2.2.5 CRSF Interface Baud Rate

The Crossfire Serial (CRSF) protocol carries RC channel data between the RP2 and the connected host ESP32 over a full-duplex UART interface. The baud rate must match on both sides. A rate of 115,200 bps was used, which is the default CRSF baud rate supported by the RP2 hardware and the standard Arduino `HardwareSerial` peripheral on the ESP32.

A 26-byte CRSF frame at 115,200 bps with 8-N-1 framing occupies approximately 2.26 ms on the wire. Because frames are sent every 20 ms (50 Hz command rate), the duty cycle is approximately 11%, leaving ample headroom for returning telemetry frames in the same UART window without frame collision.

2.2.6 Telemetry Configuration

ELRS supports bidirectional telemetry: the flight controller returns sensor data (battery voltage, received signal strength, link quality) through reserved downlink slots. Two operating modes were configured.

RC control mode. When the RP2 transmits as the control head connected to the ESP32 ground unit, the telemetry ratio is set via the `elrsV3.lua` script on the transmitter. A ratio of 1:64 was configured, allocating one downlink telemetry slot for every 64 uplink RC command packets. At a 250 Hz command update rate this yields approximately 3.9 telemetry packets per second, sufficient for battery voltage and link quality monitoring without perceptibly reducing command throughput.

Ground station serial mode. When the RP2 is connected by USB to a laptop running ground station software, the *Use as AirPort Serial device* option is enabled in the WebUI. In this mode, all received CRSF telemetry frames are forwarded as raw serial data through the module's USB-to-UART bridge at 400,000 bps, allowing ground station applications to parse and display real-time flight data without a separate RF modem.

2.3 Drone Flight Subsystem

2.3.1 Airframe and Propulsion

The airframe is a five-inch class quadrotor based on a Mark4-style five-inch frame (purchased as a complete frame kit). Four brushless outrunner motors are installed; each is rated for 1860 KV and is paired with a blade compatible with the five-inch propeller class. A five-inch propeller protection set was added (vendor listing: four-piece guard set for five-inch class props) to reduce damage during low-altitude development flights.

2.3.2 Flight Control Electronics

The autopilot uses a SpeedyBee F405 V5 series flight controller. Motor commutation is driven by a dedicated OX32 55 A, four-in-one ESC wired to the four 1860 KV outrunners. During procurement the team received *two* four-in-one ESCs from different man-

ufacturers: one shipped with the initial SpeedyBee-oriented stack bundle and the OX32 unit purchased as a separate line item. Because the brands differed and the newer OX32 hardware matched the final integration plan, the earlier ESC was taken out of the flight harness, retained as laboratory spare stock, and processed for reimbursement under the spare-equipment policy, while the OX32 module remained in the demonstrator.

An ExpressLRS compatible airborne receiver (RX24T) terminates the RF link from the ground-side RadioMaster RP hardware described in Section 2.2. The flight controller communicates with that receiver over the same CRSF serial interface convention used throughout the ELRS ecosystem.

Procurement note. Vendor invoices occasionally abbreviate bundled boards with non-specific stock codes while still shipping SpeedyBee-class hardware; the installed configuration documented here was verified on the bench and in flight, not inferred from invoice text alone.

2.3.3 Commissioning and Calibration

Before first flight the following commissioning steps were performed using ground station software connected to the SpeedyBee F405 V5 over USB.

Sensor calibration. Accelerometer calibration was carried out by placing the airframe on a flat surface and following the six-position guided routine in the ground station software; each face of the airframe was held still while the software recorded the gravity vector. Gyroscope calibration was performed with the airframe stationary on a bench to establish the zero-rate offset for each axis. The flight controller stores both calibration sets in onboard flash and reloads them at power-on.

Motor direction verification. Each motor was spun individually at minimum throttle through the ground station motor test page. Two motors were found to rotate in the incorrect direction for the chosen propeller-rotation scheme and were reversed by swapping any two of the three ESC phase wires at the ESC pads. After swapping, all four motors were confirmed to spin in the intended direction before propellers were fitted.

Flight controller configuration. The SpeedyBee F405 V5 ships with a Betaflight firmware image. The factory PID values were retained for all three attitude axes; no manual PID tuning loop was performed. This decision was informed by the fact that a five-inch class quadrotor at 1860 KV is a well-characterized platform, and the factory Betaflight defaults produced stable, flyable behavior in early hover tests without further adjustment.

2.3.4 Manual Flight Qualification

A RadioMaster Pocket transmitter was used for manual flight qualification prior to integrating the smartwatch control path. The pilot adapted to the aircraft's handling characteristics over multiple hover and forward-flight sessions, verifying stable attitude hold in angle mode. This baseline established a reference for evaluating whether the smart-

watch control path introduced any perceivable latency or authority degradation relative to conventional stick input.

3 Verification

Verification was organized in three stages: unit-level bench tests for each subsystem, followed by integration tests of the full RF link on the ground, and finally flight tests with all subsystems operating simultaneously. The Requirement and Verification Table is provided in Appendix A.

3.1 Voice Control Unit

3.1.1 Voice Recognition Latency (R1)

The VC02 recognition latency was evaluated by issuing each of the seven commands five times in a quiet indoor environment and recording the time from the end of the spoken keyword to the appearance of the “Voice CMD = X” debug line on the ESP32 serial console.

3.1.2 CRSF Frame Transmission (R2, R3)

The CRSF output was verified with a logic analyzer on the ESP32 GPIO 25 TX line. Frames were confirmed to be 26 bytes in length, carrying address byte 0xC8, frame type 0x16, and a valid DVB-S2 CRC-8. Inter-frame spacing was measured at 20 ms (± 1 ms), corresponding to a 50 Hz command rate that satisfies R3. The end-to-end software latency from VC02 UART byte reception to the first CRSF frame departure was bounded by the 20 ms `lastSend` timer interval, satisfying R2.

3.2 ELRS RF Link

3.2.1 Link Establishment

The RP2 transmitter and RX24T receiver were powered simultaneously with matching Binding Phrases compiled into both firmware images. Link establishment (green LED on RX24T) was observed within 5 s of power-on in all bench tests, consistent with the ELRS automatic binding specification.

3.2.2 Channel Fidelity

RC channel values output by the RX24T were verified via the Betaflight Configurator receiver tab. Each commanded channel position (1000, 1500, 2000 μ s for roll, pitch, yaw, and throttle; 1000 and 2000 μ s for arm) was observed to arrive at the flight controller within ± 3 μ s of the intended value, within Betaflight’s normal channel noise floor.

3.2.3 Telemetry Downlink

Barometric altitude telemetry (CRSF frame type 0x09) was received by the ESP32 and decoded correctly, with the altitude value displayed on the OLED matching the Betaflight Configurator blackbox readout to within ± 0.2 m during static bench tests.

3.3 Drone Commissioning

Accelerometer and gyroscope calibration were verified by confirming that the Betaflight Configurator attitude indicator remained stable and level ($<2^\circ$ reported tilt) when the aircraft was placed on a flat surface after calibration. Motor directions were verified by visual inspection of prop-wash direction during spin-up tests at minimum throttle, with all four motors confirmed correct before propellers were fitted.

3.4 Flight Tests

Over the course of the project Chen Renang conducted more than ten flight sessions using the RadioMaster Pocket transmitter for baseline verification and the voice control unit for demonstration flights.

Hover stability (R4).

Directional response (R5).

Landing success (R6).

Altitude telemetry (R7). The OLED displayed a valid altitude reading throughout all flight sessions in which the CRSF downlink was active.

4 Costs

5 Costs

Table 4 summarizes major purchased items for which the team holds vendor invoices, expressed in *tax-included* renminbi (RMB) exactly as printed on those invoices. Consumables such as solder appear in the miscellaneous group. Labor estimates follow the course guideline: for each partner, ideal hourly salary $\times$ actual hours worked $\times 2.5$. Replace the placeholder hours and rates below with values agreed by the team and supported by lab notebooks or time logs.

Notes: [†]Amounts split from a single invoice that paired the F405-class board with the RX24T. [‡]Seller line read “H10d10”; bundle had a four-in-one ESC from a different vendor than OX32. That ESC was retired from the flight harness (spare reimbursement); stack row is the bundled charge. If F405 V5 appears twice on invoices, drop one row in a single-airframe BOM. Two ESCs were paid for; only OX32 is installed (Section 2.3).

Machine-shop charges or university internal service fees, if any, should be added in the same manner with explicit citations to work orders.

Table 4: Major procured components (tax-included RMB totals from vendor e-invoices).

Category	Item (as used on the aircraft)	Qty	Total (RMB)
RC transmitter	RadioMaster Pocket (manual base-line flights)	1	369.00
ELRS hardware	RadioMaster RP1 and RP2 modules (ground + development)	2	198.00
Airframe	Mark4 class five-inch frame kit	1	65.80
Prop guards	Five-inch guard set (vendor: four pieces per set)	1 set	66.31
Flight computer	SpeedyBee F405 V5 flight controller [†]	1	348.99
Airborne RX	ExpressLRS RX24T receiver [†]	1	92.00
Power stage (flight article)	OX32 55 A four-in-one ESC (final installed unit after vendor change)	1	298.00
Stack bundle	SpeedyBee F405 V5 stack invoice line (includes FC plus original four-in-one ESC [‡])	1	152.99
Motors	1860 KV brushless outrunner	4	263.80
Battery	Six-cell (6S) 22.2 V, 1400 mAh class LiPo	1	168.00
Power connector	AMASS XT60 pigtail pair	1	2.75
Hardware kits	M3 screw assortments (two vendor line items combined)	2 kits	5.73
<i>Miscellaneous laboratory support</i>			
Solder	0.8 mm leaded wire spool	1	16.90
Battery charger	“T6” class charger for field packs	1	169.15
Microcontroller	STM32F103C6T6 module (invoice line item)	1	16.98

Table 5: Labor cost worksheet (placeholder—replace hours and salary assumptions).

Partner	Role focus	Hours	Rate (USD/h)	Cost (USD)
Partner A	(smartwatch firmware)	—	—	—
Partner B	(RF link / ELRS)	—	—	—
Partner C	(flight test)	—	—	—
Total labor (formula: hours × rate × 2.5)				—

6 Conclusions

This project demonstrated that a voice-command interface built on commodity offline recognition hardware can control a quadrotor drone through the full mission cycle of takeoff, directional flight, and landing without a conventional RC transmitter. The ESP32 and VC02 combination achieved the required command latency budget, the ELRS link maintained reliable packet delivery at 50 Hz across all flight sessions, and the SpeedyBee F405 V5 platform proved stable on factory PID defaults without manual tuning.

6.1 Summary of Accomplishments

The following objectives from the high-level requirements were met:

- A seven-word offline voice command vocabulary was implemented and mapped to flight-controller CRSF channel sequences with correct DVB-S2 framing and timing.
- The ELRS RF link was established using a RadioMaster RP2 reflashed as a CRSF transmitter, achieving automatic pairing via Binding Phrase without physical button binding.
- Stable hover was maintained for multiple seconds after each takeoff command, and directional pulse maneuvers produced observable displacements.
- Barometric altitude telemetry was decoded from the CRSF downlink and displayed on the OLED throughout flight, providing the ground operator with real-time situational awareness.
-

6.2 Ethical Considerations

This project involved repeated outdoor and indoor flight of a motorized aerial vehicle with rotating blades capable of causing injury. The IEEE Code of Ethics [2] requires engineers to hold the safety of the public paramount and to be honest about the limitations of their work. In adherence to these principles, all flight tests were conducted in restricted, supervised areas with bystanders kept outside the rotor arc. The voice control system incorporates a software interlock that rejects directional commands unless the takeoff sequence has already been completed, preventing unintended ground-level motor commands. Future deployment in public spaces would require regulatory compliance with local UAV operating rules and additional hardware failsafes such as a geofence or automatic return-to-home function.

The project has broader relevance in the context of accessible human-machine interfaces. Voice-driven UAV control reduces the physical skill barrier for operators with limited manual dexterity and opens potential applications in disaster response and remote inspection where operator hands may be otherwise engaged. At the same time, increased accessibility of UAV control raises societal concerns around privacy and airspace safety that future iterations of this work should address through both technical design choices and engagement with regulatory frameworks.

6.3 Future Work

Several improvements would increase the practical utility of the system.

Higher command update rate. The current 50 Hz CRSF output rate is sufficient for stable flight but is below the 250 Hz or 500 Hz rates achievable with ELRS. Increasing the rate would reduce the latency between a voice command being received and the first corresponding CRSF frame reaching the aircraft.

Continuous control modality. Replacing discrete pulse commands with proportional control—for example, by mapping the operator’s head orientation or a wrist-worn IMU to roll and pitch channels in real time—would enable more precise maneuvering and altitude hold.

Safety failsafes. A hardware kill-switch input to the ESP32 and a CRSF failsafe channel configuration on the flight controller would ensure the motors disarm if the voice link or USB power to the ground unit is interrupted unexpectedly.

Outdoor range characterization. All verification flights were conducted at short range. A systematic link-budget test at increasing distances would characterize the operational envelope of the RP2-based transmitter at the ISM_2400 regulatory domain power level.

References

- [1] ExpressLRS Contributors. ""ExpressLRS: Open Source RC Link"," Accessed: May 1, 2026. [Online]. Available: <https://github.com/ExpressLRS/ExpressLRS>
- [2] IEEE. ""IEEE Code of Ethics"," Accessed: May 1, 2026. [Online]. Available: <https://www.ieee.org/about/corporate/governance/p7-8.html>

Appendix A Requirement and Verification Table

Table 6 lists every requirement from Table 1 alongside the verification method used and the result.

Table 6: Requirement and Verification Table.

Req.	Verification method	Target	Result
R1	Time five utterances of each command; record UART byte timestamp via ESP32 serial monitor.	<1 s	<i>TODO</i>
R2	Logic analyzer on GPIO 25; measure time from VC02 byte in ISR to frame first bit on wire.	<40 ms	Pass (≤ 20 ms)
R3	Logic analyzer inter-frame period on GPIO 25 TX line.	≥ 50 Hz	Pass (50 Hz)
R4	Record hover duration after each takeoff command across all flight sessions.	≥ 5 s	<i>TODO</i>
R5	Observe drone displacement during forward/backward voice commands in open area; estimate distance by reference marker.	Observable	<i>TODO</i>
R6	Count landing attempts vs. successful disarms (motors stop, craft does not invert or crash).	100%	<i>TODO</i>
R7	Confirm OLED altitude reading is non-zero and updating during each flight session.	Continuous	<i>TODO</i>